

Backend Local Setup (FastAPI + MySQL)

1) Requirements

- Linux (Ubuntu/Debian recommended)
- Python 3.10+
- MySQL (local)
- `pip / venv`

Optional system packages (helps with builds + OpenCV):

```
sudo apt-get update
sudo apt-get install -y build-essential python3-dev libgl1
libglib2.0-0
```

2) Install MySQL locally (script)

Run `setup_mysql.sh` in your backend root:

```
Cd Backend
chmod +x setup_mysql.sh
./setup_mysql.sh
```

Verify MySQL:

```
sudo systemctl status mysql --no-pager
```

4) Create venv + install requirements

From backend root (where `requirements.txt` exists):

```
python3 -m venv .venv
```

```
source .venv/bin/activate
pip install --upgrade pip
pip install -r requirements.txt
```

5) Run the backend

Run FastAPI:

```
python -m uvicorn main:app --host 0.0.0.0 --port 8080
```

Open:

- API docs (Swagger): <http://localhost:8080/docs>
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6) Quick API checks

Add a camera (example):

```
curl -X POST "http://localhost:8080/cameras" \
-H "Content-Type: application/json" \
-d '{"rtsp_url":"rtsp://user:pass@ip:554/stream","enabled":true}'
```

Stream preview (MJPEG in browser):

- http://localhost:8080/cameras/<camera_uuid>/stream.mjpg?fps=10

Snapshot:

- http://localhost:8080/cameras/<camera_uuid>/snapshot.jpg
-

Troubleshooting

MySQL not running

```
sudo systemctl restart mysql  
sudo systemctl enable mysql
```

Access denied

```
sudo mysql -e "SELECT user, host FROM mysql.user;"
```

OpenCV import errors

```
sudo apt-get install -y libgl1 libglib2.0-0
```
